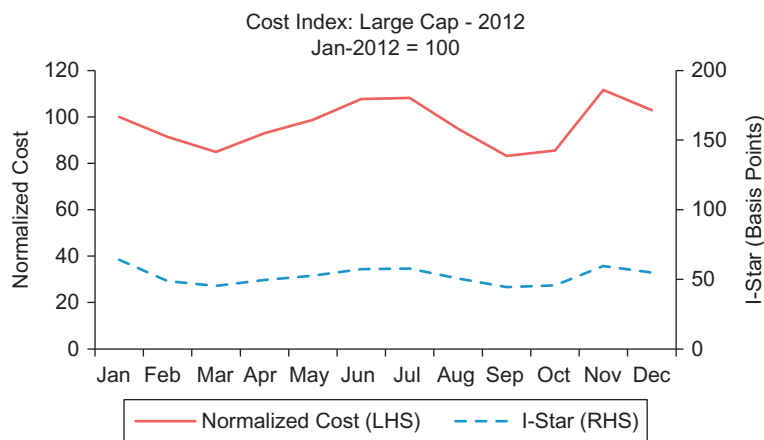




■ Figure 12.2 Normalized Large Cap Cost Index.

An alternative approach is to construct the cost index for the desired time period using the current trading cost for the specified order size and adjusting for the normalized index value. For example, suppose the manager is interested in the cost of trading an order of 20% ADV with a $POV = 25\%$ strategy. The current trading cost for this order is 51 bp, and the current and previous month's normalized index are 105 and 95 respectively. From the normalized cost index we estimate that the trading cost in the previous month was $95/105 = 0.90$ or 90% less. Therefore, the expected cost for trading 20% ADV via $POV = 25\%$ is $95/105 \times 51 \text{ bp} = 46 \text{ bp}$.

The normalized cost index provided the necessary information to recreate historical trading costs without the need to reconstruct the index for this size and strategy. Furthermore, we did not need access to historical market conditions or market impact parameters since this information was already embedded into the normalized cost index time series.

The backwards-propagation process can be generalized as follows:

$$\text{Trade Cost}(t - k, \text{Size}, \text{POV}) = \text{Current Trade Cost}(t, \text{Size}, \text{POV}) \cdot \frac{\text{Normalized Index}(t - k)}{\text{Normalized Index}(t)} \quad (12.4)$$

Where $\text{Trade Cost}(t - k, \text{Size}, \text{POV})$ is the historical cost at time $t - k$ for the specified Size and POV , and $\text{Current Trade Cost}(t, \text{Size}, \text{POV})$ is the current trading cost at time t for the identical Size and POV rate. The current trading cost is estimated via the preferred market impact model (e.g., see Chapter 5 or Chapter 11).